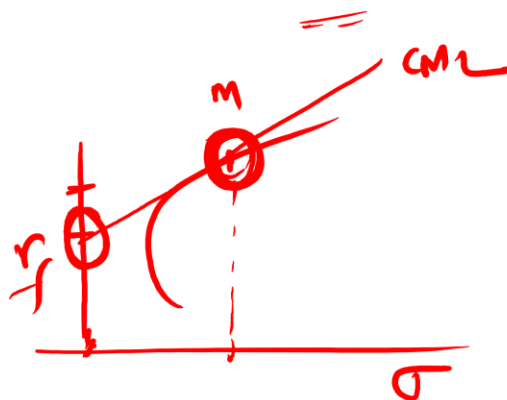


Portfolio Management (3. Factor Models)

Factor Theory



- ▶ Factors drive risk premiums ✓
- ▶ Factors can be macro economic such as growth, inflation, productivity etc.
- ▶ Factors can also be tradable investment styles such as value-growth, momentum etc.
- ▶ Factor investing refers to earning risk premium. This has two rationales
 - Economic – factors pack risk premiums as a way of compensating for the low returns during bad times associated to the factors
 - Behavioral – factors cannot be diversified away (arbitraged away)



The Factor Zoo



Macro factors

- Economic Growth ✓
- Inflation ✓
- Volatility ✓
- Productivity Risk ✓
- Demographic Risk ✓
- Political Risk ✓

Investment-Style factors

- Static factor (Market)
- Dynamic factor
 - Size ✓
 - Value ✓
 - Momentum ✓

Macro Factors – (Economic Growth)



	Large Stocks ✓	Small Stocks ✓	Govt Bonds ✓	Corporate Bonds	
				Investment Grade ✓	High Yield ✓
Means					
Full Sample →	11.3%	15.3%	7.0%	7.0%	7.6%
Business Cycles (1)					
Recessions ✓	5.6%	7.8%	12.3%	12.6%	7.4%
Expansions ✓	12.4%	16.8%	5.9%	6.0%	7.7%
Real GDP (2)					
Low	8.8%	12.2%	10.0%	9.7%	7.0%
High	13.8%	18.4%	3.9%	4.4%	8.2%
Consumption (3)					
Low	5.6%	5.6%	9.6%	9.1%	7.1%
High	17.1%	25.0%	4.4%	5.0%	8.2%
Inflation (4)					
Low	14.7%	17.6%	8.6%	8.8%	9.2%
High	8.0%	13.0%	5.4%	5.3%	6.0%
Volatilities					
Full Sample →	16.0%	23.7%	10.6%	9.8%	9.5%
Business Cycles					
Recessions	23.7%	33.8%	15.5%	16.6%	18.1%
Expansions	14.0%	21.2%	9.3%	7.8%	6.8%
Real GDP					
Low	16.9%	23.7%	12.2%	11.8%	12.1%
High	14.9%	23.7%	8.5%	7.0%	6.0%
Consumption					
Low	17.5%	23.8%	11.9%	11.6%	11.8%
High	13.8%	22.7%	8.9%	7.4%	6.6%
Inflation					
Low	15.5%	21.9%	9.6%	8.2%	7.7%
High	16.4%	25.4%	11.5%	11.1%	11.0%

Quarterly returns from 1952:Q1 to 2011:Q4

▶ Economic Growth :

Mean Return

- During recession, Equities perform poorly. Small cap stocks have larger disparities in mean returns between periods of recession and expansion.
- Govt. bonds and IG bonds perform well during recession and relatively worse during expansion.
- High yield bonds do not exhibit much difference in their mean returns.

Volatility

- All assets exhibit higher volatility during recession.

Macro Factors – (Inflation)

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Quarterly returns from 1952:Q1 to 2011:Q4

► Inflation :

Mean Return

- ❑ All assets perform poorly during high inflation
- ❑ High inflation lowers the value of the bonds in real terms. It's not so obvious why stocks perform badly during high inflation.

Volatility ✓

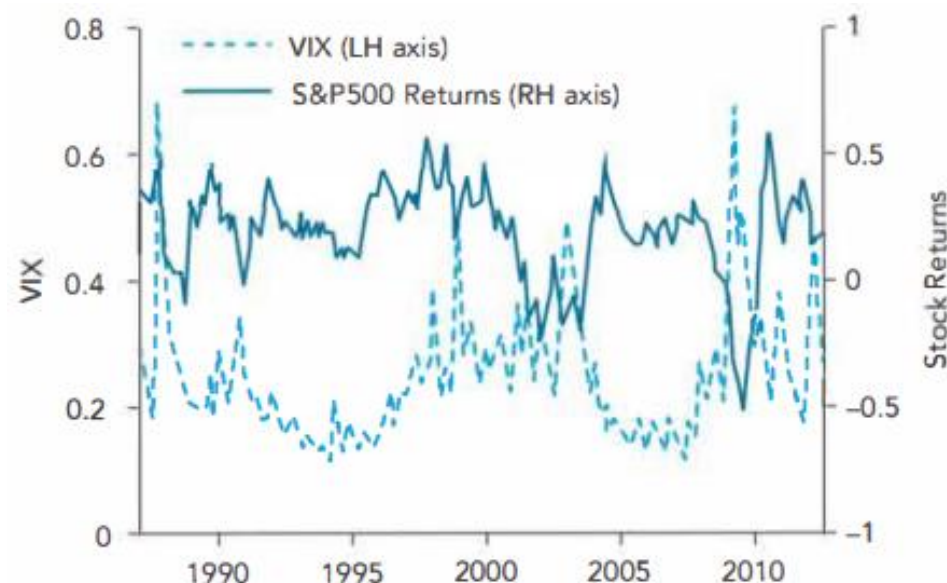
- ❑ All assets exhibit higher volatility during high inflation.

Macro Factors – (Volatility)

σ_t

Correlation Matrix

	VIX Changes	Stock Returns	Bond Returns
VIX Changes	1.00	-0.39	0.12
Stock Returns	-0.39	1.00	-0.01
Bond Returns	0.12	-0.01	1.00



- ▶ When volatility rises, stocks perform poorly
- ❑ Leverage effect – When the asset returns drop, the equity price falls increasing the D/E (leverage) ratio which makes the firm risky hence volatility rises
- ❑ Risk Return Tradeoff – Investors demand more premium for taking higher volatility (risk). So, equity prices must decline now, so future returns are high.
- ▶ Bonds are not a safe haven during periods of high volatility as evident from very low correlation.

$E(r) \uparrow$

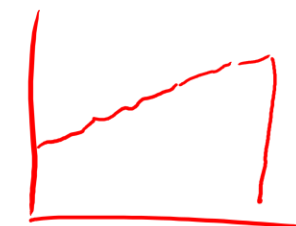
Trading Volatility

- ▶ Many assets (other than stocks) perform poorly when volatility is high.
- ▶ Investors who want to get protected from volatility can buy put options. (why not call ?)
- ▶ Investors who can afford to take volatility risk can sell protection (sell put options)
- ▶ One can also enter into other types of derivatives such as variance swaps.
- ▶ An average investor would pay to avoid volatility risk than be paid to take it on.
- ▶ During stable times, one can collect volatility premium by selling volatility because VIX volatilities are generally 2% to 3% higher than realized volatility. But during crash, there are huge losses associated with selling volatilities. This works something like insurance.
- ▶ CAPM equation $E[R_m] - R_f = \gamma \sigma_m^2$ indicates that market premium should be positively correlated with market volatility in theory, but in data the sign has been seen as time varying (negative, 0 and positive)

$V_{St} > 0$

$(-ve)$
 $(K - S)^+$
↓

$\gamma \rightarrow$ risk aversion coeff
 > 0



Other Macros – Productivity Risk

► **Productivity Shocks**

- Productivity risk refers to the real business cycle shocks (firm output)
- Stock prices rise with rise with rise in productivity and fall with fall in productivity. Correlation $\sim 48\%$ between TFP shocks and stock returns. But productivity shocks are a more long-term phenomenon, so it does not matter to short term investors

► **Demographic Risk**

- Demography risk refers to labor output shocks which is a slow-moving variable that affects long-term investors
- Labor income is earned and saved during youth and middle age, while expenses are done during old age. Cohorts progress through these stages and join previous cohorts (overlapping generation models). A demographic shock is change in the composition of the cohorts relative to other cohorts through war or disease etc.
- When retiree population increases, there is low investment in financial assets and more expenses. Assets are sold to meet those expenses. This leads to fall in asset prices.

► **Political Risk**

- Greater the political risk, higher the risk premium demanded by the investors
- It used to be a concern for only emerging markets, but after GFC it's also a concern for developed economies

Dynamic Factor Models

- ▶ In 1976, Stephen Ross introduced the first multi-factor equilibrium multifactor model, the **Arbitrage Pricing Theory (APT)**, which takes the form:

$$R_i = R_f + \beta_1 F_1 + \beta_2 F_2 + \dots + \beta_n F_n + \epsilon_i$$

- ▶ Key differences between CAPM and APT

$$\frac{E[R_i] - r_f}{1} = \beta_1 E(F_1) + \beta_2 E(F_2) + \dots + \beta_n E(F_n)$$

- ❑ **Multiple Risk Factors:** CAPM assumes a **single factor (market risk)** whereas APT allows for **multiple factors** (e.g., inflation, GDP growth, interest rates, oil prices).
- ❑ **No Market Portfolio Assumption:** CAPM assumes that all investors hold the **market portfolio**. APT does not require a market portfolio, making it **more flexible**.
- ❑ **Arbitrage-Based Pricing:** APT relies on the concept of **no arbitrage for each factor**. CAPM assumes **equilibrium pricing** based on investor behavior.

Fama-French (1993)

- Fama-French model uses three factors to explain asset returns

$$E[R_i] = R_f + \beta_1 E[R_m - R_f] + \beta_2 E[SMB] + \beta_3 E[HML]$$

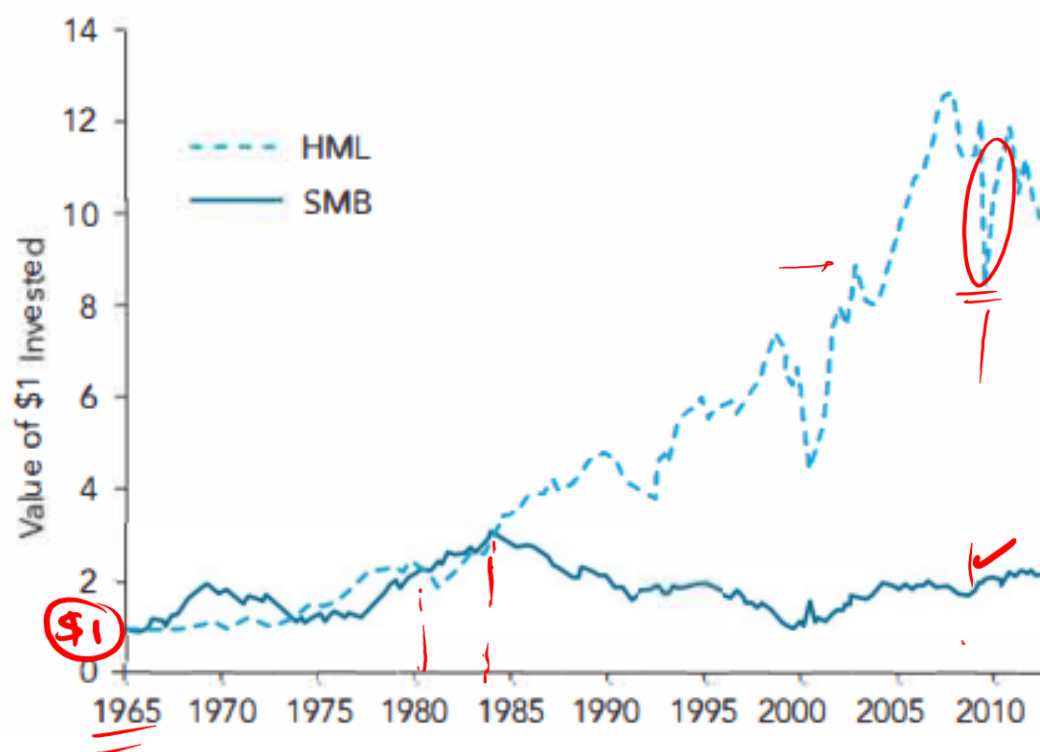
Handwritten notes and annotations:

- B/M (circled in red) with an arrow pointing to the HML term.
- P/B (circled in red) with an arrow pointing to the HML term.
- B (circled in red) with an arrow pointing to the HML term.
- M (circled in red) with an arrow pointing to the HML term.
- $\eta \times S$ (circled in red) with an arrow pointing to the HML term.
- $R_m - R_f$ (bracketed in red) with an arrow pointing to the market premium term.
- SMB (circled in red) with an arrow pointing to the Size term.
- HML (circled in red) with an arrow pointing to the Value-Growth term.
- Arrows pointing to the β_1 , β_2 , and β_3 coefficients.

- SMB : small – big (captures the premium of small cap stocks over the large cap stocks)
- HML : high – low (captures premium of high B/M stocks over low B/M stocks). It's also referred to as value – growth strategy. High B/M stocks are undervalued stocks, also called as value stocks which has more potential to give high returns.
- An average stock has only market exposure. It does not have a size or value tilt.
- A value stocks will have positive beta for HML factor. A small cap stock will have positive beta on SMB factor. These premiums add on to market premium.

Performance of SMB and HML

- ▶ Fama-French model uses three factors to explain asset returns



- ▶ **Size factor** was discovered in 1981. Right after the discovery, the size factor disappeared.
 - Two reasons are stated
 - 1) Rosenthal's 'file drawer' problem, i.e., only 5% of results which are positive are published
 - 2) Investors bid up the prices of the small stocks until the effect disappears
- ▶ **Value premium** is robust and has stood the test of time.

Theories for Value Premium

► Rational theories of Value premium

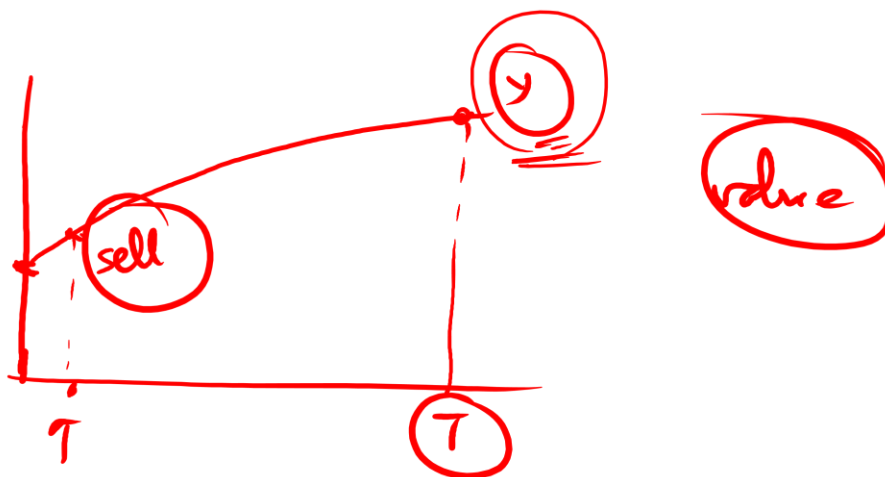
- Managers use real investment options to increase firm value. They exercise these options optimally when the market returns are low. This non-linear effect is not captured in a linear model like CAPM. This gives rise to value premium. →
- Value stocks are capital intensive industries. During bad times they are not flexible enough to pivot to new opportunities and bear losses. This makes the value stock riskier, hence the long-term premium.

► Behavioral theories of Value premium

- This theory is around over-optimism and over-exploitation. Value stocks are not inherently risky as postulated by rationale theories, rather the growth is simply underestimated.
- On the other hand, growth stocks are subject to excessive optimism.
- Psychological biases : *loss-aversion* , *mental accounting* . Investors suffer losses more than rejoice equivalent gains. The value stock is the one with low market value as a result of some prior bad performance. This is conceived by the investor as risky, hence expect premium.

Values in other Asset Classes

- ▶ The concept of value can be extended to other asset classes. We have to remember that value (low price) is synonymous with high yield.
- ▶ In Equities, it's called value-growth strategy
- ▶ In Fixed Income, it's called riding the yield curve (buy long term bonds and sell before maturity)
- ▶ In FX, it's called carry (long currency with higher interest rates, short currency with lower interest rates)



Momentum Investing

- ▶ Momentum investing refers to buying stocks which have gone up (winners) in the past (say 6 months) and shorting stocks with the lowest returns (losers) in the same period. The central idea is that winners will continue to grow, and losers will continue to fall.
- ▶ Value investing is a negative feedback strategy, it's inherently stabilizing, because one would buy the undervalued stocks and therefore the price will go up.
- ▶ Momentum investing is a positive feedback strategy which basically means buy the winners which are already going up. This is ultimately destabilizing and subject periodic crashes.
- ▶ Sometimes, the factor WML is added to the Fama-French model

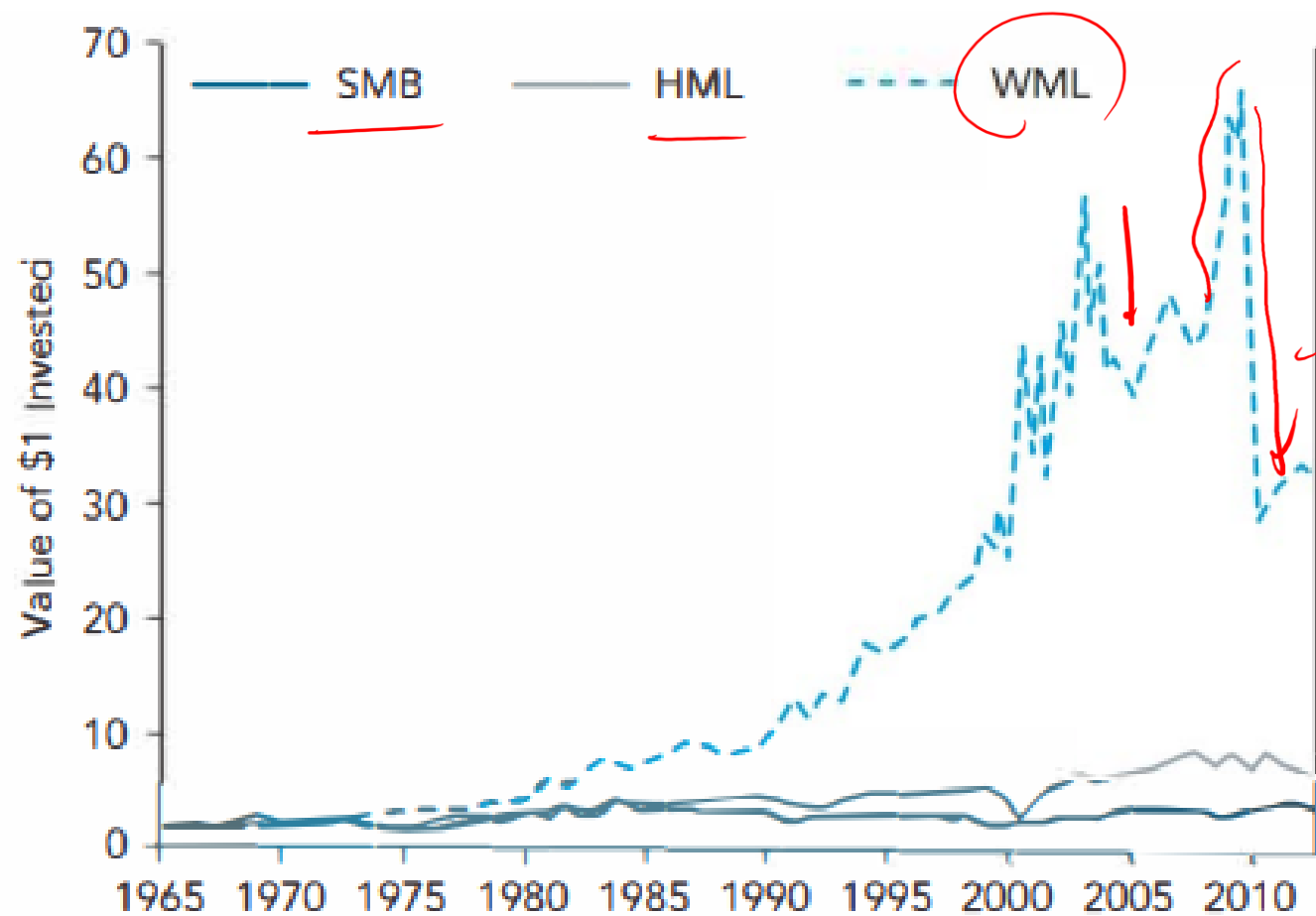
WML

↑
↓ ↓

$$\mathbb{E}[R_i] = R_f + \beta_1 \mathbb{E}[R_m - R_f] + \beta_2 \mathbb{E}[SMB] + \beta_3 \mathbb{E}[HML] + \beta_4 \mathbb{E}[WML]$$

(market premium)
(Size)
(Value-Growth)
(Momentum)

Performance of SMB, HML, WML



Momentum Risks



► Tendency towards crashes –

- Despite very good average return, momentum strategy is subject to crashes

► Govt. Risks ✓

- Govt interferes with natural market dynamics. Govt. puts a floor to declining stocks, sometimes bails them out. Stocks rebound, causing the strategies to break.

► Business Cycle

- Research suggests that momentum is affected by business cycles and the state of the stock market. .



Generating Momentum

- ▶ There are two most widely behavioral theories that describe how momentum arises

Overreaction

- Informed overconfident investors overestimate their abilities to forecast a firm's cashflows. They have a biased self-attribution when they are successful, that it must be so because of their skill. But when things don't go their way, they attribute it to bad luck. So, when these investors hear positive signals about a firm, they attribute the good performance to their own skill leading to overconfidence. They overreact and push the prices up beyond their fundamental values, causing momentum

Underreaction

- Some Investors, new-watchers underreact to the signals. This gets partially incorporated to the price. But the investors learn and act with a delay pushing the prices up subsequently.

